

# OS BACKUP SYSTEM DOCUMENTATION

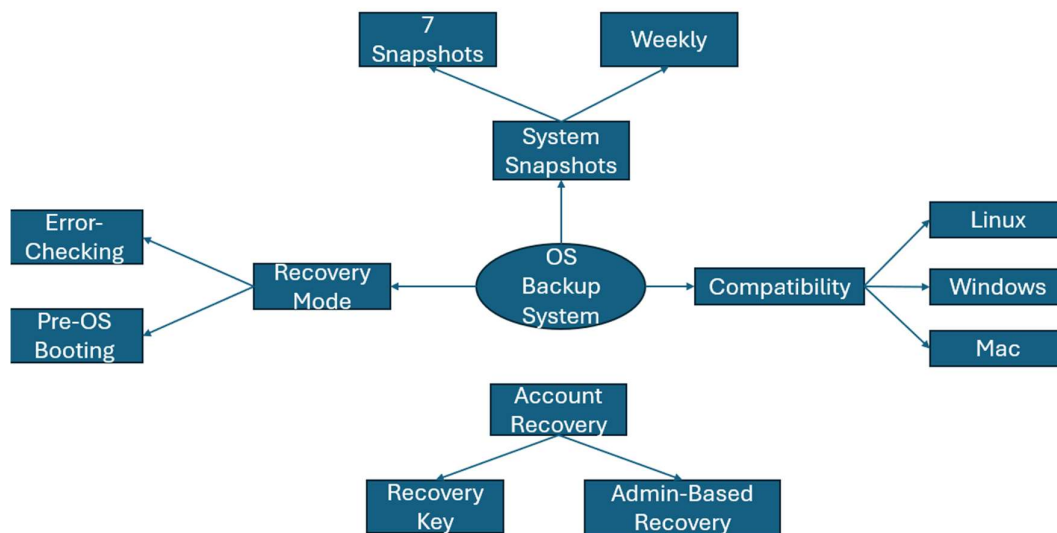
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## Introduction

This OS Backup System is designed to provide current existing operating system and model for how recovery and backups should be designed. It is also designed to provide further resilience and redundancy in operating systems in general. This OS Backup System was developed as currently recovery modes and redundancy systems are at a good level but could use reinforcement in some areas. This OS Backup System will explain recovery features and mode, the recovery workflow and the expected trade-offs of this backup system.

This OS Backup System cannot promise perfect recovery, resilience or redundancy within an operating system. However, this OS Backup System can promise multiple reliable and resilience features to recover a device. This is because it will cover recovery features across accounts, encryption, data breaches and software errors. This OS Backup System is also compatible with existing operating systems and would require software updates to change. Make sure to adapt this OS Backup System based on what operating system its designed for.



The spider diagram above shows a visual layout of the OS Backup System. This visual layout involved 2-3 features of each concept of the OS Backup System. This diagram does not show all features of recovery mode, system snapshots, and account recovery. More details can be found in the sections below.

## OS Backup System Recovery Mode

- Every operating system needs a recovery mode just in case of technical issues within the operating system.
- There are 4 areas that this OS Backup System's recovery mode focuses on.

- The first area is activation of recovery mode.
- Recovery mode can be entered either if the user initiates it or if a major technical issue has been detected.
- The user can initiate recovery mode via the settings and authenticating via their admin password or account password (if they are on their admin account)
- Once recovery mode has been activated, the device will restart and load into the recovery environment instead of the usual operating system.
- The user will be presented with the options to recover via snapshots, run an error-check scan, reinstall OS via USB or cloud storage (if internet connection is available).
- This is important as recovery mode must be able to work without relying on the usual operating system's functionality.
- The second area is full system snapshots
- These snapshots are created by the operating system to create a save point where the operating system is clean and are stored in a separate recovery partition.
- The operating system creates 7 of these with an overwrite policy of weekly with deleting the oldest created snapshot.
- These snapshots are created if there isn't an active threat detected by anti-virus software and are verified via a verification hash created by the OS.
- In recovery mode, these snapshots can then be used to restore the OS to a clean state with minimal data loss after they are verified.
- The third area is error-checking
- This is where the recovery environment checks the device for firmware, kernel-level and software-based errors.
- The operating system then presents the results of the scan in the format What (What was the error?) → Why (Why did it mostly likely occur?) → Options (What the user can do to fix the error?)
- The fourth area is reinstalling the OS via USB or cloud storage.
- If there is internet connection, the recovery environment can request an new clean install from a verified cloud server hosting the clean version.
- If there is no internet connection, the recovery environment can allow booting from an USB to reinstall the operating system.
- This is the final and nuclear option which will result in complete data loss of the device.
- These four areas provide resilience, redundancy and automated recovery throughout the backup system.
- Make sure to adapt the recovery mode options to have terminal commands for Linux-based operating systems or for power users.

## **OS Backup System Account Recovery and Export Policy**

- It is also important that users can recover operating system accounts and export their data from the operating system.
- For account recovery, there are two ways for users to recover their accounts.
- The first way is admin-based recovery.
- This is where the user logs into their admin account to reset the password for standard user accounts they may have.
- The user must authenticate via their admin account's password before they can reset other users' password to prevent recovery-based attacks.
- However, this approach only works if the user has standard and admin separation on their device which is currently only mainly available on Windows and Linux.
- The second way is direct account recovery
- This is where the user can either use a recovery key, reset their account via email (if connected) or answer security questions.
- This recovery key is showed to the user, and they are asked to store the key somewhere securely separate from their device.
- It is also important to remember that email recovery and security questions are weaker recovery mechanisms and email recovery can raise privacy concerns.
- Each operating system may use a different recovery mechanism, but I would recommend recovery key as it is the most secure option.
- For export policy, there are 2 main methods the user should be able to use.
- The first method is creating an OS snapshot directly.
- This is where the user creates the OS snapshot and saves it to a separate device, cloud storage or a physical drive.
- This can then be used during recovery mode via USB as it would be able to support snapshots if they are verified via the verification hash.
- It is important to remember that this acts as final option before completely reinstalling the operating system.
- The second method is downloading a clean version of the operating system from the cloud servers via a settings option.
- This allows the user to have a direct, verified and built in option for moving a clean version of the operating system onto USB.
- This is important because it is free and reduces the risk of compromise where the OS may install a malicious version of the operating system via the cloud server.
- Make sure to adapt this export policy and these account recovery methods based on the specific operating system.

## **OS Backup System Recovery Workflow**

- This recovery workflow covers the process from first device boot to full recovery via USB on a windows machine using this system.
- First, the user boots up their new laptop with Windows 11 installed and ready to go.
- Second, Windows 11 immediately uses windows defender to scan for any threats on the device just in case.
- Third, Windows 11 creates the snapshot and verification hash for the recovery partition.
- Fourth, the user enables device encryption and is asked to save the recovery key on a separate device.
- Windows 11 allows the user to save it on their device (with warning), on the cloud, or on a connected physical drive.
- Fifth, the user saves the recovery key in cloud storage and on an old HDD they had laying around.
- Sixth, Over the next 5-6 weeks, 7 snapshots have been created by the operating system using the same workflow.
- Seventh, one day the user boots up their laptop when the OS runs into an error.
- The user is then prompted into recovery mode to fix the error.
- Eighth, the user runs an error-check scan to find out the error.
- The results are that the error is Boot Configuration Data (BCD), likely caused by corrupted boot configuration data. The user's options are to allow the automated process to manually fix the boot configuration or restore from a snapshot.
- The user chooses to allow the automated process to fix the boot configuration data corruption.
- 4-5 minutes later, the user can boot back to the operating system with no data loss.
- Ninth, a few months later, the user's device is compromised by ransomware via an SMB vulnerability as the user uses the protocol for work.
- Windows Defender detects the ransomware via its real-time protection and prevents access to core systems files. It then removes it successfully.
- Tenth, the user then initiates recovery mode after entering their admin password causing their device to restart and then enter the recovery environment.
- The user then clicks on the restore from snapshot option where they are allowed to choose which snapshot to restore from.
- The snapshots are verified via the verification hash before the restoration process begins.
- After the selected snapshot is verified, the user can restore the operating system to a clean state with minimal data loss.

## **OS Backup System Trade-offs**

- **This OS Backup System has trade-offs just like any other system I have built so far.**
- **There may also be additional implementation challenges developers and engineers can face.**
- **I have identified 6 main trade-offs within this OS Backup System.**
- **The first trade-off is snapshot infection.**
- **This trade-off exists because malware may be able to infect the snapshot on the recovery partition.**
- **This means that an attacker can have persistent access even after the user restores from a snapshot.**
- **This trade-off can be managed through AV scan before creation, verification hash to check for tampering and offering multiple other options.**
- **The second trade-off is admin lockout.**
- **This trade-off exists because if the user is locked out from their admin account they can lose access to their data.**
- **This trade-off can be managed via recovery key, cloud reset (privacy concern), or security questions to allow the user to regain access to their account.**
- **The third trade-off is storage overhead**
- **This trade-off exists because the device must store 7 full system snapshots and the recovery environment.**
- **This trade-off can be managed by a user-configurable overwrite policy, separate partition and more lightweight recovery features on budget devices.**
- **The fourth trade-off is complexity**
- **This trade-off exists because device error and recovery processes can be difficult and confusing for the user to navigate.**
- **This trade-off can be managed by user-friendly interface, multiple automated options, clear and consistent reports and terminal for power users.**
- **The fifth trade-off is compatibility**
- **This trade-off exists because every operating system handles recovery differently due to their design or Kernel architecture.**
- **This trade-off can be managed by adapting this recovery system to the specific OS, incremental updates and using targeted recovery systems based on user base.**
- **The sixth trade-off is maintenance cost**
- **This trade-off exists because developers must constantly update the error-checking process and recovery process to ensure security and usability.**
- **This means that developers and engineers can be exhausted constantly updating and fixing bugs or errors within the recovery processes.**
- **This trade-off can be managed through rotating schedules; incremental updates and an error-handling focus on errors that prevent snapshot-based recovery.**